Deep Learning and the Human Brain: Clarifying the Analogy

CS4152 — Deep Learning and Neural Networks

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Outline

Why People Compare Deep Nets to the Brain

- Simplified neuron model resembles logistic regression unit.
- Forward + backward propagation are hard to explain intuitively.
- Media popularized the "brain-like" narrative.
- Historically useful for intuition, but increasingly inaccurate.

A Questionable Analogy

- Biological neurons are vastly more complex than ANN units.
- Neural computation involves biochemical, electrical, and temporal dynamics.
- ANN units: simple algebra + activation functions.
- Superficial resemblance; fundamentally different mechanisms.

Learning Mechanisms Differ

- Brain:
  - Unknown full learning algorithm.
  - Likely non-differentiable and non-layered.
  - Synaptic plasticity, spiking, local rules.
- Deep Learning:
  - Backpropagation
  - Gradient descent
  - Global error signals

Why the Analogy Is Less Useful Today

- Deep nets are designed for flexible function approximation.
- Many architectures (e.g., transformers) have no biological analogue.
- Biological plausibility is not a design goal.
- Some inspiration remains in computer vision (e.g., convolutional layers).

Deep Learning: A Functional, Not Biological Paradigm

- Neural networks: compute complex $x \rightarrow y$ mappings.
- Brain analogy: oversimplified, often misleading.
- Recommended: emphasize mathematical and computational viewpoints.
- Final takeaway: deep learning brain simulation.

Key Takeaways

- Historical analogy between ANNs and the brain is shallow.
- Biological neurons and learning are much more complex.
- Deep learning operates through backprop, not biological learning.
- Better to treat deep nets as powerful function approximators.